

Nanocellulose-based foam morphological, mechanical and thermal properties in relation to hydrogel precursor structure and rheology

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ABSTRACT

Foams were prepared from nanocellulose-based hydrogel precursors using a freeze-drying process. The work mainly aims at investigating the relationships between the mechanical and thermal properties of foams and the rheological properties of their hydrogel precursors, which were characterized in a previous paper. The structure of foams was characterized by SEM and confocal microscopy, their elasticity by compression tests, and their thermal conductivity by hot strip as well as transient pulsed techniques. A strong correlation was shown between the elastic properties of foams and those of their hydrogel precursors, and a minimum thermal conductivity was shown to appear at a cellulose volume fraction corresponding to a transition in viscoelastic properties of hydrogels. Results suggest that foams and hydrogels share common microstructural features, which makes it possible to tune the mechanical and thermal properties of foams by tuning the rheological properties of their hydrogel precursors.

1. Introduction

Bio-based aerogels and foams, mainly prepared from polysaccharides, are relatively new highly porous eco-materials which have attracted more and more attention in the last decade (Alnaief, Alzaitoun, García-González, & Smirnova, 2011; Fischer, Rigacci, Pirard, Berthon-Fabry, & Achard, 2006; Ganesan et al., 2018; Gavillon & Budtova, 2007; Rudaz et al., 2014; White, Budarin, Luque, Clark, & Macquarrie, 2009; Zhao, Malfait, Guerrero-Alburquerque, Koebel, & Nyström, 2018). For example, cellulose aerogels (Budtova, 2019) were shown to be of particular potential interest for thermal insulation applications in buildings (Jiménez-Saelices, Seantier, Cathala, & Grohens, 2017; Nguyen et al., 2014), exhibiting thermal performances close to commercial materials from petrochemical sources (Illera, Mesa, Gomez, & Maury, 2018; Muthuraj, Jimenez-Saelices, Grohens, & Seantier, 2018). Besides, significant efforts were made in recent years in order to improve the mechanical properties of such cellulose-based materials (De France, Hoare, & Cranston, 2017; Lavoine & Bergström, 2017; Martoia et al., 2016).

The thermal properties of this new generation of cellulose-based foams and aerogels are very promising, and super-insulating cellulose aerogels, with thermal conductivity lower than $0.024 \text{ W.m}^{-1} \cdot \text{K}^{-1}$, were

reported (Ayadi et al., 2016; Kobayashi, Saito, & Isogai, 2014). These remarkable thermal properties can be obtained due to the Knudsen effect (Notario et al., 2015), because of a highly porous structure enabling the trapping of air in very small cavities. Actually, the thermal heat transfer through aerogels and foams results from three main contributions: the heat conduction through the solid skeleton, the heat conduction through the gas, and the radiative heat transfer contribution. Structures of low density, with very small pore sizes, lead to a low solid conduction as well as a low gas conduction, responsible for very good insulation properties.

Cellulose aerogels and foams are generally prepared following a two-step protocol. The first step consists in getting a hydrogel. CNF hydrogels can be formed at very low volume fractions, and their structural, rheological properties and mechanical stability result mainly from the competition between inter-fibrillar H bonds and electrostatic interactions, but also from the effect of CNF entanglements (De France et al., 2017; Lavoine & Bergström, 2017). In a second step, the hydrogel precursor is dried using either a supercritical-drying or a freeze-drying process in order to remove the liquid phase within the hydrogel, resulting in the formation of CNF aerogels or foams (Chen, Yu, Li, Liu, & Li, 2011; Korhonen, Kettunen, Ras, & Ikkala, 2011; Miettunen et al., 2014; Pääkkö et al., 2008; Zu et al., 2016). One of the most challenging

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issue is to try to preserve the fragile solid cellulose network during the drying process. From that point of view, the supercritical drying process has the advantage to avoid capillary forces during drying, and therefore to avoid the pore collapse phenomenon, resulting in the formation of aerogels exhibiting very good thermal insulation (Pierre & Pajonk, 2002). The second method, classically used to form aerogels and foams, consists in freezing followed by sublimation of the frozen solvent. It is a rather simple, easily adaptable method, but which may induce structural changes in the solid network, due to ice templating, affecting the final thermal and mechanical properties (2013, Deville, 2010).

Thermal and mechanical properties of cellulose-based aerogels and foams were shown to depend on a great variety of factors and parameters controlling their structure, from the source of cellulose and its chemical and mechanical treatment to the processing parameters and conditions (Chen et al., 2014; Lavoine & Bergström, 2017; Martoia et al., 2016). From that point of view, it is highly important to analyze and try to elucidate the links between the properties of foams or aerogels and those of their hydrogel precursors. The present work aims at getting through this difficult issue, by studying the relationships between a) the morphological, mechanical and thermal properties of CNF foams and b) the structure and rheological properties of their hydrogel precursors, which were investigated in a previous work (Fneich, Ville, Seantier, & Aubry, 2019). Finally, we would like to point out that the objective of the work is not to get CNF foams with the best mechanical and thermal properties using an up-scalable optimized process, but rather to investigate whether some structural signature of hydrogels can be found in foams prepared from these precursors.

2. Experimental section

2.1. Materials

TEMPO-oxidized wood cellulose nanofibers (CNF) used in this work were supplied by EMPA (Swiss Federal Laboratories for Materials Science and Technology), in the form of a 1 wt% aqueous suspension. CNFs were prepared from wood pulp which was Elementally Chlorine Free bleached, then treated by a TEMPO-mediated oxidation process, followed by a mechanical treatment. The lignin and hemicellulose content of CNF has not been characterized in the present work, but both the Elementally Chlorine Free bleaching process and the TEMPO mediated oxidation treatment are known to remove most of lignin and hemicellulose from the CNF surface (El Bakkari, Bindiganavile, Goncalves, & Boluk, 2019; Laine & Stenius, 1997). This nanocellulose was characterized in a previous work: using conductivity measurements, CNFs were shown to have a carboxylate content of 1.12 mmol/g; using XRD measurements, a crystalline content of about 70 % was estimated; finally, using atomic force microscopy, CNFs were shown to exhibit an average length of about 300 nm, an average diameter of about 2.5 nm, and a persistence length of about 280 nm (Fneich et al., 2019).

2.2. Hydrogel precursors

Foams studied in this work were prepared from CNF hydrogels. Starting from 1 wt% CNF suspensions, hydrogel precursors were prepared at volume fractions of 0.3 %, 0.4 %, 0.45 %, 0.5 %, 0.6 % in deionized water and at volume fractions of 0.1 %, 0.25 %, 0.3 %, 0.4 %, 0.5 %, 0.6 %, and 0.65 % with 50 mM NaCl. The difference in volume fraction range with and without salt is mainly due to the difference in sol/gel transition of CNF suspensions with and without salt. Indeed, a hydrogel network was shown to form at a lower CNF fraction in the presence of salt (Fneich et al., 2019). The structural and rheological properties of these soft materials were thoroughly studied in a previous paper (Fneich et al., 2019). One of the most interesting and original result of this work was the existence of a transition at a CNF volume fraction of 0.4 % in deionized water, defining two different viscoelastic regimes, characterized by two different power-law exponents of the

elastic modulus versus volume fraction curve. This behavior was attributed to changes in the structure of the junctions of the hydrogel 3-dimensional network: increasing the CNF content, an increasing influence of inter-fibrillar entanglements was shown, resulting in stronger junctions at higher volume fractions. Besides, adding 50 mM NaCl was shown to modify the extent and behavior of the hydrogel linear viscoelastic response.

2.3. Foams preparation

CNF foams studied in this work were prepared from their hydrogel precursors by a freeze-drying method, following a previously described protocol (Abdelwahed, Degobert, Stainmesse, & Fessi, 2006). CNF hydrogels were poured into molds, and put, without any stirring, into a MDF-C8V1 ultra-low temperature freezer, at a temperature of - 80 °C, that was proved to be optimum in a previous work, for 24 h (Jiménez-Saelices et al., 2017). Then, molds were put into a Christ Alpha 1–2 LD Plus freeze-dryer for solvent removal. Freeze-drying was performed at a condenser temperature of - 50 °C, and carried out under vacuum, at a pressure below 0.1 mbar. All samples remained in the freeze-dryer for 5 days.

2.4. Morphological characterization

A Hitachi-S-3200 SEM (Scanning Electron Microscopy), with an accelerating voltage of 15 kV, was used to observe the CNF foams morphology. All samples were metallized prior to observation. Confocal microscopy was performed as an additional characterization technique in order to observe directly the 3-dimensional structure, using a ZEISS confocal microscope, with a laser beam of 405 nm.

2.5. Mechanical characterization

Uniaxial compression tests were performed on foams using a TA Instruments - DMA 2980 mechanical analyser, at room temperature. Foam samples used for mechanical characterization are in cylindrical shape, with a length to diameter ratio of 2/3. They are put between 2 parallel plates, the bottom one being fixed, and the upper one moving at a constant speed of 1 mm/min. All compression tests were performed up to 100 % deformation, in controlled conditions ($T = 21$ °C and relative humidity = 50 %). The linear part of the nominal compression stress – engineering strain curves was used to determine the compressive Young's modulus of foams.

2.6. Thermal characterization

Two different techniques were used to characterize the thermal properties of foams: the hot strip technique and the transient pulsed technique. Foam samples with different dimensions were used: $2 \times 2 \times 0.4$ cm³ for the hot strip technique, and $8 \times 8 \times 1$ cm³ for the transient pulsed technique.

2.6.1. Hot strip technique

The effective thermal conductivity, λ_{eff} , of CNF foams was determined using the steady state hot strip technique. This classical technique allowed to determine the conductance, K , of the samples, which is related to their effective thermal conductivity:

$$K = A\lambda_{eff} + B \quad (1)$$

where A and B are two constants, which were determined during a preliminary calibration step using materials of known thermal conductivity.

2.6.2. Transient pulsed technique

In order to study the various heat transfer modes through foams,

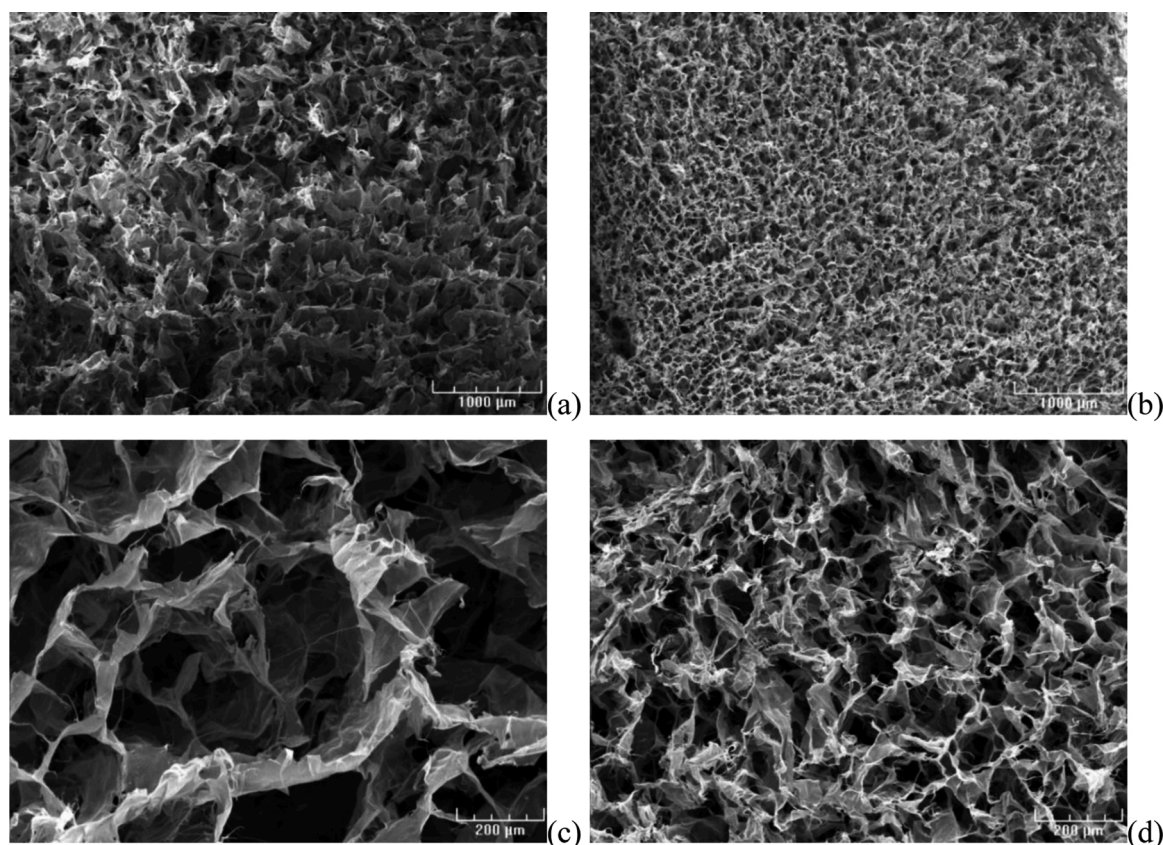


Fig. 1. SEM micrographs of foams at a volume fraction of 0.3 % (left) and 0.6 % (right) at different length scales.

including solid and gas conduction as well as radiation, a specific device, based on a pulsed technique and dedicated to the thermal characterization of bio-aerogel and foams, was used. The device and model used to analyze the thermal properties have been previously described in detail (Pierre, Jimenez-Saelices, Seantier, & Grohens, 2017). This technique allows to decouple the conductive from the radiative contributions to the heat transfer, by considering the radiative transfer as a pure diffusion process characterized by a radiative conductivity, λ_r , adding to the thermal conductivity, λ_c .

3. Results and discussion

3.1. Morphological properties

3.1.1. SEM observations

The SEM micrographs at different length scales of the morphology of foams at a volume fraction of 0.3 % and 0.6 % are plotted in Fig. 1. These two CNF fractions were chosen to illustrate the effect of cellulose content on the foam morphology, but also because they are on both sides of the volume fraction (0.4 %) marking the transition between two different viscoelastic behaviors of the hydrogel precursors, attributed to two different network structures (Fneich et al., 2019).

Fig. 1 shows that the pores of the 0.6 vol% CNF foam are not only significantly smaller, but also look more homogeneous in size and shape, than those of the 0.3 vol% one. Even if the network structure of the hydrogel precursor is obviously strongly affected by the ice-templating process, the increasing morphological homogeneity of foams, when increasing CNF content, was attributed to the increasing effect of CNF entanglements in the hydrogel network (Fneich et al., 2019). More precisely, we suggest that parts of the CNF skeleton in the hydrogel network are not destroyed during the freeze-drying process, so that both materials (foams and hydrogels) share common “bricks”, that is common microstructural features. The increasing number of CNF

entanglements, when increasing CNF fraction, makes these structural “bricks” less breakable and deformable, therefore leading to a better morphological foam homogeneity at higher CNF fractions. This explanation is indirectly confirmed by SEM observations of foams prepared from hydrogels in the presence of 50 mM NaCl, as presented in Fig. S1. Actually, SEM microscopy shows that adding salt at 0.3 % or 0.6 % CNF volume fraction has the same qualitative effect on the structure as adding further CNF particles, because of the increase of CNF entanglements induced by electrostatic screening (Fneich et al., 2019).

3.2. Confocal microscopy observations

In order to precise the foam morphology at increasing CNF fractions, confocal microscopy observations were performed. This technique has two advantages compared to SEM: it does not need any sample preparation and allows to get a 3-dimensional characterization of the foam structure. Fig. 2 shows typical 2-dimensional (projected) and 3-dimensional confocal observations of 0.3 vol% and 0.6 vol% CNF foams.

Using 2D micrographs like those in Fig. 2a and b, an average pore diameter, calculated over about 50 pores, was determined with the aid of SigmaScan Pro 5 software. At a 0.3 % CNF volume fraction, the average pore diameter is about 120 μm , whereas it is about 80 μm at 0.6 % CNF volume fraction. However, 3D micrographs like those in Fig. 2c and d reveal the great complexity of the topology of the open cell foam morphology, which cannot be simply characterized in terms of pore diameters, but would require a detailed analysis using topological descriptors. This will be done in a future work exclusively devoted to the study of CNF aerogels and foams morphology.

3.3. Mechanical properties

In order to characterize the linear mechanical properties of CNF foams, the compression Young's modulus, E_c , as a function of the CNF

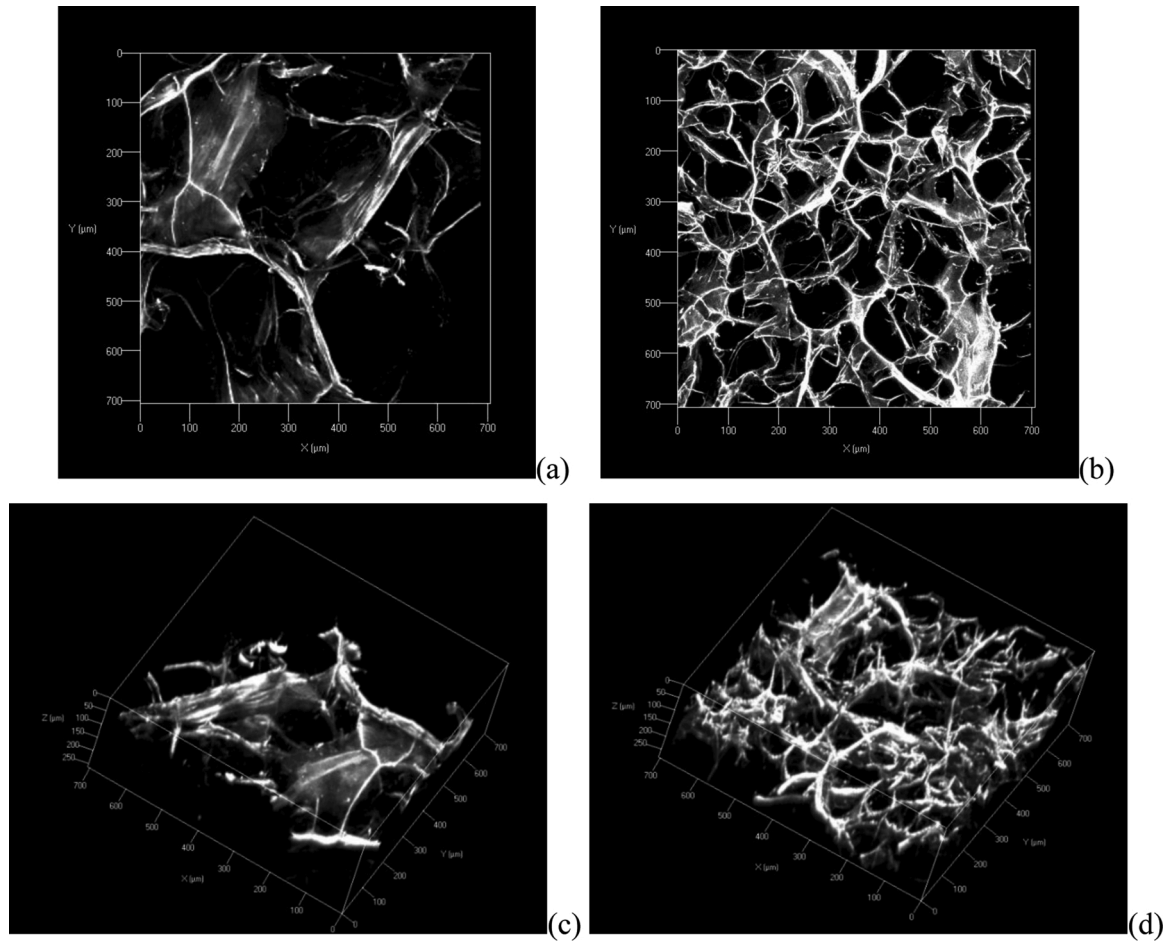


Fig. 2. 2D (projected) and 3D confocal observations of foams at a volume fraction of 0.3 % (left) and 0.6 % (right).

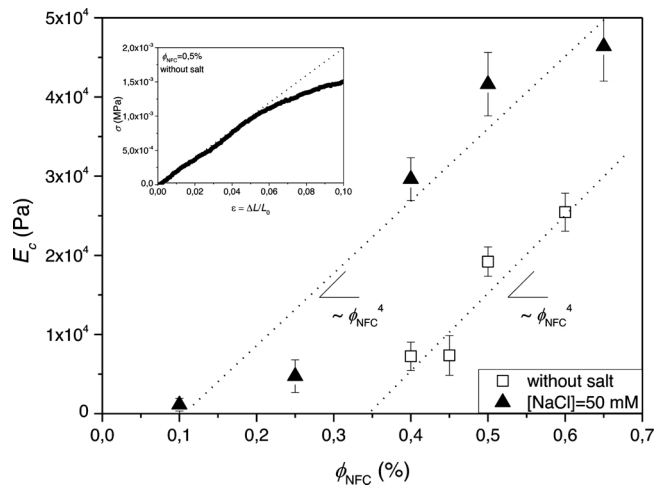


Fig. 3. Compression Young's modulus of foams as a function of CNF volume fraction, with 50 mM NaCl and without salt.

volume fraction was plotted in Fig. 3, prepared with 50 mM NaCl and without salt. The insert in Fig. 3 shows the linear fit of the stress-strain compression curve used to determine E_c .

Results plotted in Fig. 3 show that the compression Young's modulus of foams, prepared with or without salt, increases with the CNF volume fraction following the same power-law dependence:

$$E_c \sim \phi^4 \quad (2)$$

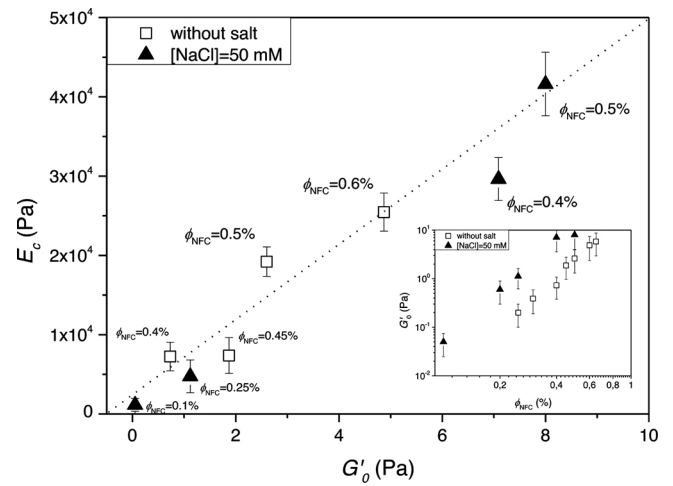


Fig. 4. Compression Young's modulus of foams as a function of storage modulus of their hydrogel precursors.

However, foam elasticity is significantly increased in the presence of 50 mM NaCl. This is in agreement with the increase of CNF entanglements, due to electrostatic screening, which enhances the cohesive (and therefore the elastic) properties of the solid cellulose network, as discussed in our previous work (Fneich et al., 2019).

In order to study the possible relationships between the linear elastic properties of foams and those of their hydrogel precursors, the

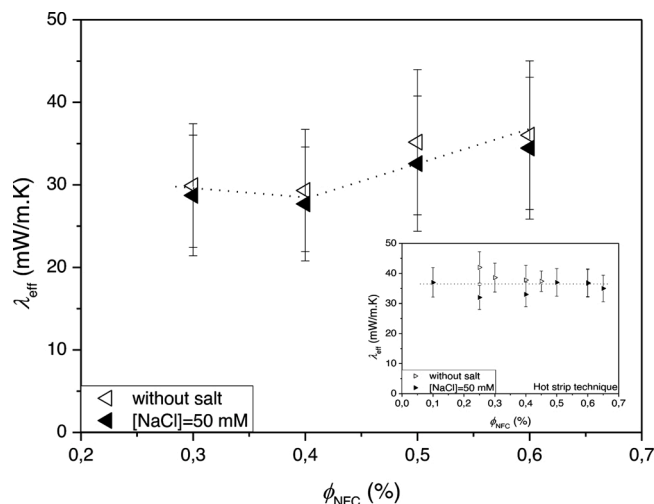


Fig. 5. Thermal conductivity of foams as a function of CNF volume fraction, with 50 mM NaCl and without salt.

compression Young's modulus, E_c , of foams was plotted as a function of the elastic (storage) modulus, G'_0 , of their hydrogel precursors (Fneich et al., 2019) in Fig. 4. The G'_0 versus CNF volume fraction, with 50 mM NaCl and without salt, was plotted in insert of Fig. 4.

Quite interestingly, results in Fig. 4 clearly show that the linear elastic properties of foams correlate with the linear mechanical properties of their hydrogel precursors. Such a correlation is in qualitative agreement with the assumption of common structural features shared by CNF foams and their hydrogel precursors. These common “bricks” are elastically active and seem to govern the response of both foams and hydrogels to small deformations.

3.4. Thermal properties

The effective thermal conductivity, λ_{eff} , measured by the transient pulsed technique, is plotted as a function of CNF volume fraction, for foams prepared with and without NaCl, in Fig. 5. The experimental results from the hot strip technique are presented in the insert.

Results in Fig. 5 first show that the effective thermal conductivity of all foams tested lies between 25 and 40 $\text{mW}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, depending on the CNF volume fraction, but also slightly on the experimental technique used to determine the thermal properties. Hence, the results show that the foams studied in this work, which contain a very low cellulose content, can be considered as rather good thermal insulators: their thermal conductivity is comparable to that of cork or glass foam (Cuce, Cuce, Wood, & Riffat, 2014). Yet, cellulose-based foams with thermal conductivities 2–3 times lower have already been prepared (Fricke, 1986). Besides, results show that adding salt slightly decreases the effective thermal conductivity of foams, which has already been observed in the literature for cellulose-based aerogels (Chen et al., 2014; Jiménez-Saelices et al., 2017; Kobayashi et al., 2014). This effect can be attributed to the screening effect of salt that increases CNF interactions, which leads to the formation of smaller pores, and thus to the reduction of the gas contribution to the effective thermal conductivity.

More interestingly, the results obtained by the transient pulsed technique suggest the existence of a CNF volume fraction of 0.4 %, marking the transition between two regimes. At low volume fractions, a regime where the thermal conductivity slightly decreases with increasing CNF content, due to the reduction of the gas conduction contribution; at higher volume fractions, a regime where the thermal conductivity increases with CNF content, due to the increase of the solid conduction contribution. It is worth pointing out that this transition volume fraction in thermal properties of foams is exactly that observed in the linear rheological properties of the hydrogel precursors (Fneich

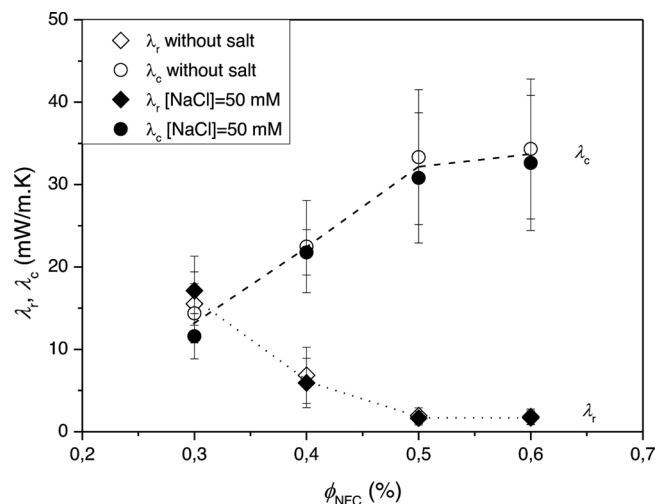


Fig. 6. Thermal conductivity and radiative conductivity of foams as a function of CNF volume fraction, with 50 mM NaCl and without salt.

et al., 2019).

In order to investigate the relative contribution of (solid and gas) conduction and of radiation to the global heat transfer, we used the transient pulsed technique. Fig. 6 shows the thermal conductivity, λ_c , and the radiative conductivity, λ_r , as a function of CNF volume fraction, for foams prepared with and without salt.

Results in Fig. 6 show that both conductive and radiative contributions mainly depend on the CNF volume fraction, since conductivities with 50 mM NaCl or without salt are quite similar. The radiative contribution decreases when increasing CNF volume fraction, as the pore diameter decreases, whereas it is the opposite for the conductive one, as the solid content increases, at least up to a CNF volume fraction of 0.5 %, beyond which both contributions seem to level off. In view of the results plotted in Fig. 4, this suggests that a delicate compromise between conductive and radiative contributions is required in order to get a minimum effective thermal conductivity (at 0.4 % CNF volume fraction).

4. Conclusion

The morphological, mechanical and thermal properties of cellulose-based foams obtained by freeze-drying from previously characterized CNF hydrogels were investigated. The focus was put on the existence of relationships between properties of foams and properties of their hydrogel precursors. The results suggest that both foams and hydrogel precursors share common elastically active microstructural features, as revealed by the strong correlation of the linear elastic properties of foams with those of their hydrogel precursors. At last, the existence of a minimum effective thermal conductivity for foams was shown to occur at a CNF volume fraction corresponding to the transition observed in the linear rheological properties of their hydrogel precursors. Finally, the overall results suggest that, for given drying process and conditions, it is possible to tune and optimize the mechanical and thermal properties of cellulose-based foams by tuning the rheological properties of their hydrogel precursors.

CRediT authorship contribution statement

Fatima Fneich: Investigation, Writing - original draft. **Julien Ville:** Validation, Visualization. **Bastien Seantier:** Resources, Methodology. **Thierry Aubry:** Project administration, Supervision, Conceptualization, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.carbpol.2020.117233>.

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